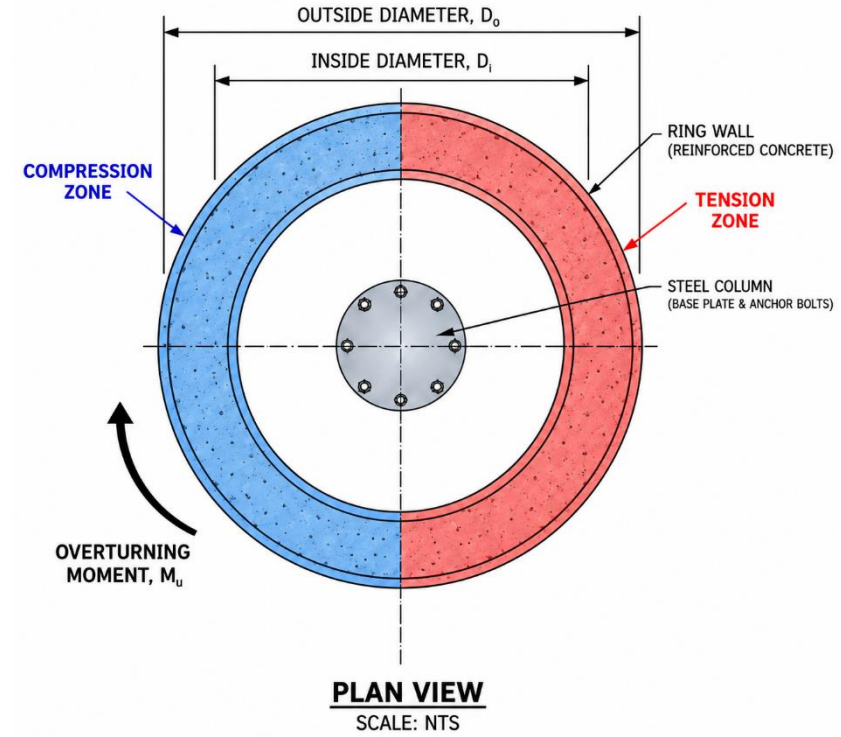
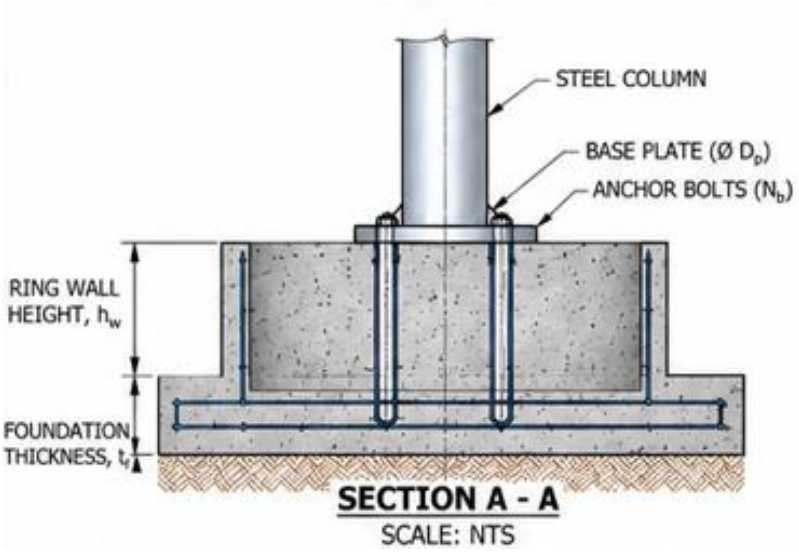
SEGMENT 3 – FOUNDATION DESIGN

100,000 Gallon Single-leg Spheroidal Water Tank

Parameter	Symbol	Value	Unit	Source
Factored axial load (LRFD)	P_u	1653.556	kip	Main Tank Design (Predefined)
Factored overturning moment (LRFD)	Mu	2903.3789	kip-ft	Main Tank Design (Predefined)
Factored shear	V_u	53.43	kip	Main Tank Design (Predefined)
Tank Cone Diameter	D_{cone}	21'-00"	ft	Main Tank Design (Predefined)
Ring-wall centerline diameter	D_{CL}	21 ft (assumed)		User Input / Preliminary Foundation Geometry
User Input / Preliminary Foundation Height	H_w	9'	ft	User Input
Ring-wall thickness	t_w	2	ft	User Input
Annular footing radial width	B_f	6.5	ft	User Input / Preliminary Foundation Geometry
Footing thickness	h_f	3	ft	User Input / Preliminary Foundation Geometry
Concrete unit weight	γ_c	150	Pcf	User Input / Standard Concrete Unit Weight
Base plate	D_p	30	in	Base Plate Design (Predefined)
Anchor bolts	N_b	4'-1.5"	in	Anchor Design (Predefined)
Concrete strength	f'_c	4.0	ksi	User Input / Foundation Specifications
Reinforcing steel yield strength	F_γ	60	ksi	User Input / ASTM A615 Grade 60
Anchor-bolt embedment depth	h_{ef}	As specified in anchor bolt design		Anchor Design (Predefined)
Allowable net soil bearing pressure	q_a	4.0	kips	User input from Geotechnical Report
Soil unit weight	γ_s	125	pcf	User input from Geotechnical Report
Active earth pressure coefficient	K_a	0.30		User input from Geotechnical Report



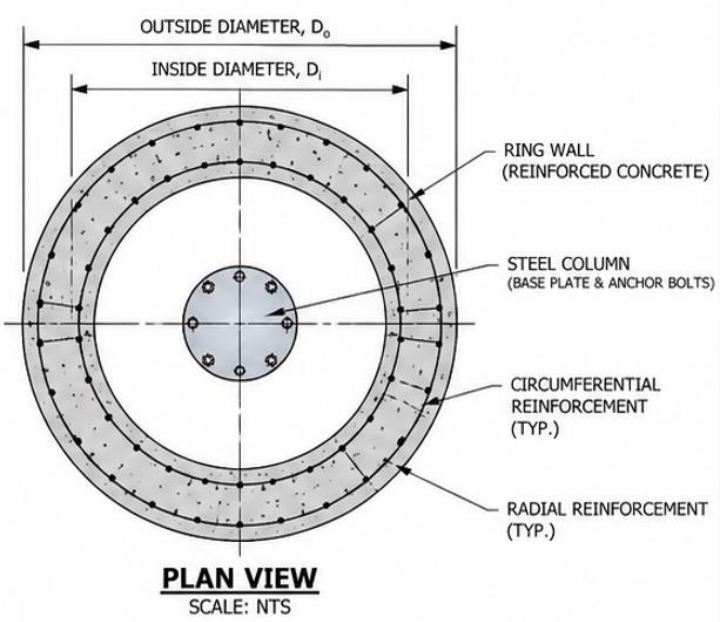


2. Geometry of Ring Wall

$$D_o = D_{CL} + t_w = 21 + 2 = 23 \text{ ft}$$

Note: users should choose the thickness of the ringwall (based on PIP STE03020 Guidelines for Tank Foundation Designs the minimum ringwall Thickness shouldn't exceed less than 12inch or 1feet).

$$D_i = D_{CL} - t_w = 21 - 2 = 19\ ft$$



Outer radius:

$$R_o = \frac{23}{2} = 11.50 \text{ ft}$$

Inner radius:

$$R_i = \frac{19}{2} = 9.50 \text{ ft}$$

Ring-wall cross-sectional area:

$$A_w = \frac{\pi}{4}(D_o^2 - D_i^2)$$

$$A_w = \frac{\pi}{4}(23^2 - 19^2)$$

$$A_w = 131.95 \text{ ft}^2$$

Ring-wall volume:

$$V_w = A_w H_w$$

$$V_w = 131.95(9) = 1{,}187.55 \text{ ft}^3$$

Using concrete unit weight of 150 pcf:

$$W_w = \frac{1{,}187.55(150)}{1000}$$

$$W_w = 178.13 \text{ kip}$$

Table 2. Minimum Inside Edge Distance

Tank Diameter D (ft)	Minimum Inside Edge Distance L (inches)
D ≤ 80	4
80 < D < 150	5
D ≥ 150	6

Reference: PIP STE03020, Tank Foundation Design Guidelines, applicable edition; project-defined geometry

Step 3. Circumferential Vertical Load Distribution

For an idealized circular support with uniform circumferential stiffness, the vertical line reaction can be represented as

$$w(\theta) = \frac{P_u}{2\pi R_{CL}} + \frac{M_u}{\pi R_{CL}^2}\cos\theta$$

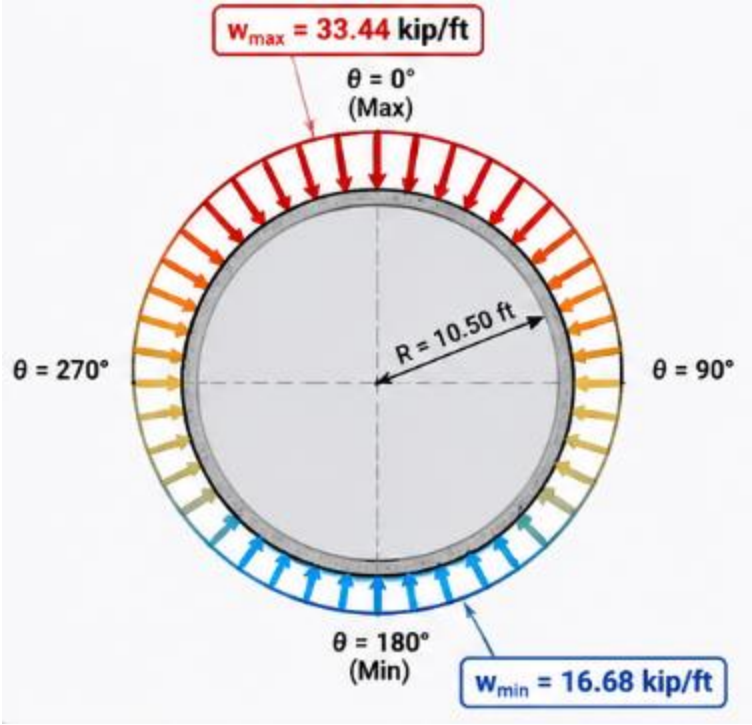
$w(\theta) =$ *vertical line reaction around the ring, kip/ft*

$P_u =$ *factored axial load*

$M_u =$ *factored overturning moment*

$R_{CL} =$ *ring − wall centerline radius*

$\theta =$ *circumferential position*



3.1. Average axial line reaction:

$$w_0 = \frac{1{,}653.556}{2\pi(10.5)}$$

$$w_0 = 25.06 \text{ kip/ft}$$

3.2. Moment-induced reaction amplitude:

$$w_M = \frac{M_u}{\pi R_{CL}^2}$$

$$w_M = \frac{2903.3789}{\pi(10.5)^2}$$

$$w_M = 8.38 \text{ kip/ft}$$

3.3. Factored ringwall self-weight per unit circumference

$$C = 2\pi R_{CL} = 65.973 \text{ ft}$$

Unfactored self-weight per unit circumference:

$$w_{sw} = \frac{178.13}{65.973} = 2.70 kip/ft$$

Or the same way we can find For a 1-ft circumferential segment:

$$V_w = (2)(9)(1) = 18 \text{ ft}^3$$

$$W_w = 18(0.150) = 2.70 \text{ kip/ft}$$

The characteristic self-weight is:

$$W_w = 2.70 \text{ kip/ft}$$

For an illustrative 1.2D load factor:

$$w_{sw,u} = 1.2(2.70) = 3.24 kip/ft$$

This assumes the ringwall self-weight is excluded from the tank design reactions.

3.4 Maximum vertical line load:

$$w(\theta) = w_P + w_{sw,u} + w_M \cos\theta$$

$$w_{max} = 28.30 + 8.38$$

$$w_{max} = 36.68 \text{ kip/ft}$$

3.5. Minimum vertical line load:

$$w_{min} = w_P - w_M$$

$$w_{min} = 28.30 - 8.38$$

$$w_{min} = 19.92 \text{ kip/ft}$$

Note: The circumferential vertical reaction distribution is an idealized structural load model used to determine the vertical line reaction around the ringwall due to the factored axial load P_u and overturning moment M_u. The model assumes that the entire ringwall circumference remains under compression.

Geotechnical design parameters, including allowable soil bearing capacity, soil unit weight, groundwater conditions, and settlement criteria, shall be obtained from the project-specific geotechnical report. Soil bearing pressure, sliding resistance, overturning stability, and overall foundation adequacy shall be evaluated separately in accordance with the applicable geotechnical recommendations and design criteria. The software shall not assume default soil properties or consider the circumferential line reaction alone as verification of foundation adequacy.

Reference: Elastic circular-ring equilibrium; ACI 318-19 §6.1, Structural Analysis, for the structural analysis framework

Step 4. Ring-wall axial compression check

Using the 2-ft wall thickness:

$$A_g = t_w L = (2.00)(1.00) = 2.00 \text{ ft}^2 = 288 \text{ in}^2$$

The maximum factored axial force for the 1-ft segment is:

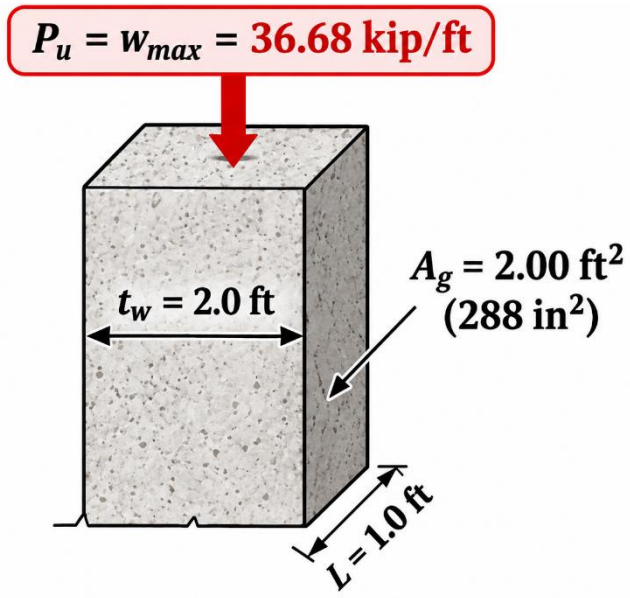
$$P_u = w_{\max} L = 33.44(1.00) = 33.44 \text{ kip}$$

$$\sigma_{\max} = \frac{P_u}{A_g}$$

$$\sigma_{max} = \frac{36.68(1000)}{288} \; ; \; \sigma_{max} = 127.4 \, psi$$

Without the additional wall self-weight:

$$\sigma_{u,\max} = 116.1 \text{ psi}$$



Note: These are average axial stresses over the assumed gross section

Ref: ACI 318-19 Sections 11.5.1 and 11.5.2.

Step 5. Vertical reinforcement check

Based on the existing design, drawing 66 #4 vertical bars on each face.

Area of one #4 bar:

Where:

$$A_b = 0.20 \text{ in}^2$$

Circumferential spacing:

$$s_v = \frac{\pi(21)(12)}{66}$$

$$s_v = 12.0 \text{ in}$$

For a 12-in-wide wall strip:

$$A_{s,v} = 2(0.20) = 0.40 \text{ in}^2$$

Vertical reinforcement ratio:

$$\rho_v = \frac{0.40}{288}$$

$$\rho_v = 0.001389 = 0.139\%$$

For the assumed low-in-plane-shear condition, ACI 318-19 Table 11.6.1 specifies:

$$A_{s,\min} = 0.0012(288) = 0.3456 \text{ in}^2$$

$$0.40 > 0.3456$$

Preliminary PASS for the assumed minimum reinforcement provision.

Reference: ACI 318-19 Table 11.6.1 and Sections 11.6–11.7. This result is conditional on satisfying the table's low-in-plane-shear applicability condition.

Step 6. Horizontal hoop reinforcement check

The drawing specifies eight #4 horizontal hoop bars on each face.

The exact vertical bar elevations must be taken from the reinforcement drawing. For a preliminary calculation only, assume 8 equal tributary spaces over 108 in:

$$s_h = \frac{108}{8} = 13.5 \text{ in}$$

Note: This is average tributary spacing, not necessarily the actual maximum spacing. Eight bars distributed between the extreme bar elevations could have a different maximum center-to-center spacing.

The approximate horizontal reinforcement ratio is:

$$\rho_h = \frac{2(0.20)}{24(13.5)}$$

$$\rho_h = 0.001235 = 0.1235\%$$

$$A_{sh,\min} = 0.0020(24)(13.5) = \ 0.648 \text{ in}^2$$

$$0.40 < 0.648$$

Preliminary PASS for the assumed minimum reinforcement provision, so we are going to select minimum As.

The required maximum spacing for two #4 bars is:

$$s_{h,\max} = \frac{2(0.20)}{0.0020(24)}$$

$$s_{h,\max} = 8.33 \text{ in}$$

Note: This is not a final reinforcement selection. If the in-plane shear demand exceeds the threshold in ACI 318-19 Section 11.6.1, the more demanding requirements of Section 11.6.2 may govern, including a minimum transverse ratio of 0.0025. The global Vu=53.43 kip must be resolved into the actual ringwall in-plane shear before selecting final horizontal reinforcement.

Step 7. Lateral soil pressure acting on the ringwall

Assume:

- Level backfill.
- Active earth pressure is mobilized.
- No surcharge.
- No groundwater pressure.
- Full 9-ft retained soil height.

$$p_a(z) = K_a \gamma_s z$$

$$p_a(z) = 0.30(125)z = 37.5z \text{ psf}$$

At the bottom:

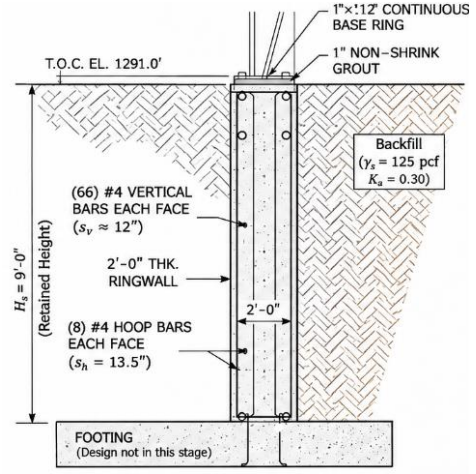
$$p_{a,\max} = 37.5(9) = 337.5 \text{ psf}$$

The lateral force per 1-ft circumferential strip is:

$$F_a = \frac{12}{K_a} \gamma_s H_s^2$$

$$F_a = \frac{12}{(0.30)}(125)(9)^2$$

$$F_a = 1.519 \text{ kip/ft}$$



The resultant acts:

$$y = \frac{H_s}{3} = 3 \text{ ft}$$

above the bottom of the retained soil height.

Conditional local cantilever moment:

$$M_a = F_a \frac{H_s}{3}$$

$$M_a = 1.51875(3)$$

$$M_a = 4.556 \text{ kip-ft/ft}$$

$$F_{a,u} = 1.6(1.51875) = 2.430 \text{ kip/ft}$$

$$M_{a,u} = 1.6(4.55625) = 7.290 kip - ft/ft$$ (For factored lateral earth pressure condition.)

8. Load eccentricity

For a 24-in-thick rectangular strip, the middle-third eccentricity limit is:

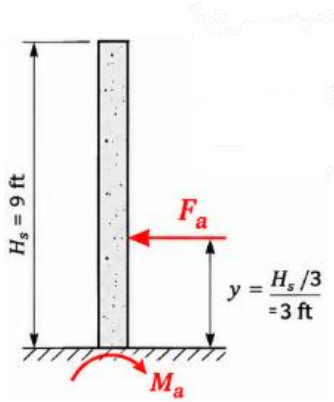
$$e_{\text{lim}} = \frac{t_w}{6} = \frac{24}{6} = 4\ inch$$

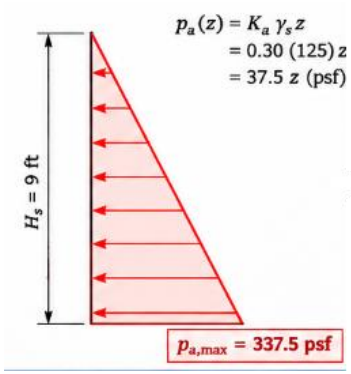
For the illustrative cantilever model, using the factored earth-pressure moment and axial load including ringwall self-weight:

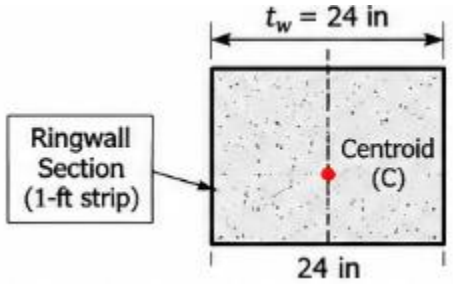
$$e = \frac{M_{a,u}}{P_{u,\max}}$$

$$e = \frac{7.290}{36.68} = 0.1987 \text{ ft} = 2.385 \text{ inch}$$

$2.385 < 4.0$ in The illustrative resultant lies within the middle third.







9. Ringwall slenderness Check

The geometric height-to-thickness ratio is:

$$\lambda_h = \frac{l_c}{h} = \frac{108}{24}$$

$$\lambda_h = 4.50$$

For a 12-in-wide rectangular strip, the radius of gyration about its weak axis is:

$$r = \frac{h}{\sqrt{12}} = \frac{24}{\sqrt{12}} = 6.928 \text{ in}$$

Assuming k=1.0:

$$\frac{kl_c}{r} = \frac{108}{6.928} = \mathbf{15.59}$$

Design status: The final slenderness condition cannot be selected from the current drawing alone. The top restraint provided by the tank shell/base ring and the base restraint must be established. Do not automatically assign k=1.0

10. Axial compression strength

$$P_n = 0.55f'_cA_g\left[1-\left(\frac{kl_c}{32h}\right)^2\right]$$

$$P_n = 0.55(4)(288)\left[1-\left(\frac{108}{32(24)}\right)^2\right]$$

$$P_n = 583.48 \text{ kip}$$

$$\phi P_n = 0.65(583.48) = 403.70kip$$

$$\text{DCR} = \frac{36.68}{379.26}$$

$$\textbf{DCR} = \mathbf{9.67\%}\ \boldsymbol{Preliminary\ axial\ compression\ strength\ PASS}$$

Reference: ACI 318-19 Sections 11.5.2, 11.5.3 and 21.2.2.

Note: The axial strength is adequate for the stated preliminary load case, provided the simplified wall method's eccentricity and boundary-condition requirements are met.

If the wall is actually braced top and bottom, the applicable lower k value may produce a higher capacity. The software should select k based on the actual support configuration rather than automatically using 2.0.

11. Circumferential hoop force due to soil pressure

A circular ringwall can resist radial soil pressure through circumferential membrane action. For an idealized circular ring of radius \(R\), the circumferential membrane force per unit wall height is:

$$N_\theta(z) = p_a(z)R$$

At the base:

$$N_{\theta,\max} = 0.3375(10.5)$$

$$N_{\theta,\max} = 3.544 \text{ kip/ft of height}$$

The corresponding average circumferential compressive stress is:

$$\sigma_\theta = \frac{N_{\theta,\max}}{t_w}$$

$$\sigma_\theta = \frac{3.544}{2}$$

$$\sigma_\theta = 1.772 \text{ ksf}$$

$$\sigma_\theta = 12.3 \text{ psi}$$

Note: This idealized membrane model applies to approximately uniform radial pressure around the circular wall. It does not represent one-sided pressure, localized tank loads, or circumferential bending. The force is compressive for inward external pressure. The software must separately consider any outward pressure that could create hoop tension.

12. Ringwall out-of-plane shear strength

The retained-soil pressure produces an out-of-plane shear demand on the assumed vertical cantilever strip.

$$V_u = F_{a,u} = 2.430 \text{ kip/ft}$$

For a 1-ft-wide strip, the effective depth:

$$d = 24 - 2 - 0.25$$

$$d = 21.75 \text{ in}$$

The vertical reinforcement area per strip is:

$$A_s = 0.40 \text{ in}^2$$

Longitudinal reinforcement ratio:

$$\rho_w = \frac{A_s}{b_w d}$$

$$\rho_w = \frac{0.40}{12(21.75)}$$

$$\rho_w = 0.001533$$

For the preliminary case, assume the provided shear reinforcement is below the applicable minimum shear reinforcement requirement and use the applicable ACI 318-19 Table 22.5.5.1 expression for that condition:

$$V_c = \left[8\lambda_s\lambda\rho_w^{1/3}\sqrt{f'_c} + \frac{N_u}{6A_g}\right] b_w d$$

For normal-weight concrete:

$$\lambda = 1.0$$

Size-effect factor:

$$\lambda_s = \sqrt{\frac{2}{1 + d/10}}$$

$$\lambda_s = \sqrt{\frac{2}{1 + 21.75/10}}$$

$$\lambda_s = 0.794$$

Axial compression contribution:

$$\frac{N_u}{6A_g} = \frac{36.68(1000)}{6(288)} = 21.23\ psi$$

Substituting:

$$V_c = \left[8(0.794)(1.0)(0.001533)^{1/3}\sqrt{4000} + 21.23\right](12)(21.75) = 17.62\ kip$$

$$\phi V_c = 0.75(17.62) = 13.22\ kip$$

Demand-to-capacity ratio:

$$DCR_V = \frac{2.430}{13.22}$$

$$DCR_V = 0.184 = 18.4\%\ Preliminary\ out - of - plane\ soil - pressure\ shear: PASS$$

Reference: ACI 318-19 Sections 11.5.5 and 22.5.5.

13. Combined axial-flexural strength

This calculation evaluates the local vertical ringwall strip under the simultaneous preliminary axial compression and soil-pressure bending demands.

Section dimensions:

$$b = 12\ \text{in}, \qquad h = 24\ \text{in}$$

$$A_{s1} = A_{s2} = 0.20 \text{ in}^2$$

Distance from each face to bar center:

$$d' = 2 + 0.25 = 2.25 \text{ in}$$

$$d = 24 - 2.25 = 21.75 \text{ in}$$

$$P_u = 36.68 \text{ kip}$$

$$M_u = 7.290 \text{ kip-ft} = 87.48 \text{ kip-in}$$

$$f'_c = 4{,}000 \text{ psi}, \quad f_y = 60{,}000 \text{ psi}, \quad E_s = 29{,}000{,}000 \text{ psi}$$

For $f'_c = 4{,}000$ psi:

$$\beta_1 = 0.85$$

$$a = \beta_1 c$$

$$C_c = 0.85 f'_c b a$$

$$\epsilon_s = 0.003\left(\frac{c - d_s}{c}\right)$$

$$f_s = \min\left(f_y, \max\left(-f_y, E_s \epsilon_s\right)\right)$$

The neutral-axis depth is iterated until the section's nominal axial force and design axial strength correspond to the applied axial demand.

$$c \approx 1.69 \text{ in}$$

$$a = 0.85(1.69)$$

The tensile strain at the far-face reinforcement is approximately:

$$\epsilon_t = 0.0357$$

Since: $\epsilon_t > 0.005$, the section is tension-controlled for this assumed flexural loading condition, with: $\phi = 0.90$

$$P_n = 40.76 \text{ kip}$$

$$M_n = 60.09 \text{ kip-ft}$$

Design moment strength:

$$\phi M_n = 0.90(60.09)$$

$$\phi M_n = 54.08 \text{ kip-ft}$$

$$\phi M_n \geq M_u$$

$$54.08 \geq 7.290$$

$$DCR_M = \frac{7.290}{54.08}$$

$$DCR_M = 13.48\%\ Preliminary\ local\ axial - flexural\ check: PASS$$

Reference: ACI 318-19 Sections 11.5.2 and 22.4.

14. ACI axial-flexural (P-M) interaction check

Parameter: b = 12in, h = 24in, Ag = 288in^2, f'c = 4ksi, fy = 60ksi, Es = 29000ksi, β1 = 0.85, Reinforcement = 1 # 4 bare at each face, As = 0.20, Cc = 2in, d' =2.25, $P_u = 36.68$ kip, $M_u = 7.29$ kip-ft

$$\rho = \frac{A_s}{A_g} = \frac{0.40}{288} = 0.001389$$

Reference: ACI 318-19 Sections 22.2 and 22.4; Section 21.2.2 for strength-reduction factors.

For each neutral-axis depth c:

$$a = \beta_1 c = 0.85c$$

$$C_c = 0.85 f'_c ba$$

$$\varepsilon_s = 0.003\left(\frac{c - d_s}{c}\right)$$

$$f_s = \min\left(E_s \varepsilon_s, f_y\right)$$

$$F_s = A_s(f_s - 0.85 f'_c)$$

$$P_n = C_c + \sum F_s$$

$$M_n = \left| C_c\left(\frac{h}{2} - \frac{a}{2}\right) + \sum F_s\left(\frac{h}{2} - d_s\right)\right|$$

Representative neutral-axis depths; capacities are per 1-ft strip.

C (in)	P_n (kip)	M_n (kip-ft)	$\emptyset P_n$ (kip)	$\emptyset M_n$ (kip-ft)
2	55.19	72.43	49.67	65.19
4	133.65	134.45	120.29	121.01

6	206.28	181.90	185.65	163.71
8	276.76	217.78	249.08	196.00
10	346.12	242.92	267.96	188.07
12	415.48	258.24	283.12	175.97
16	559.95	254.73	363.96	165.57
20	703.40	212.74	457.21	138.28
24	845.27	132.72	549.43	86.27

$$(P_u, M_u) = (36.68,\ 7.290)$$

$$(\phi P_n, \phi M_{n)} = (36.68 \text{ kip}\ ,\ 54.08 \text{ kip-ft})$$

Therefore:

$$\frac{M_u}{\phi M_n} = \frac{7.290}{54.08}$$

$$DCR_{P-M} = 0.135 = 13.5\% \textbf{.... PASS}$$

References

- ACI 318-14, Building Code Requirements for Structural Concrete — concrete strength, reinforcement, shear, development, and detailing.
- AWWA D100-21, Welded Carbon Steel Tanks for Water Storage — tank structural design and support considerations.
- PIP STE03020, Guidelines for Tank Foundation Designs — tank foundation design procedures.
- ASTM A615 Grade 60 — reinforcing steel material specification.
- Geotechnical report for the actual tank site — soil unit weight, lateral earth pressure coefficients, groundwater, bearing, and soil resistance.